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Decision Support

A MILP model to design hybrid wind–photovoltaic isolated rural electrification projects in developing countries

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ABSTRACT

Electrification systems based on the use of renewable energy sources are a suitable option for providing electricity to isolated communities autonomously. Wind and hybrid wind–photovoltaic (PV) systems are increasingly getting attention. To electrify scattered communities, designs that combine individual systems and microgrids have recently proven advantageous. In this paper we present a mathematical programming model to optimize the design of hybrid wind–PV systems that solves the location of the wind–PV generators and the design of the microgrids, taking into account the demand of the consumption points and the energy potential. The criterion is the minimization of the initial investment cost required to meet the demand. The proposed hybrid model is tested with realistic size instances and results show the instances are efficiently solved. Moreover, the model is applied to real case studies in Peru; obtained results verify that the hybrid model efficiently finds solutions that significantly reduce costs.

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1. Introduction

At present, over 1.5 billion people lack access to electricity (IEA, 2009) with this particularly affecting rural areas in developing countries (Kanagawa and Nakata, 2008). The conventional strategy for increasing access to electricity is by extending the national electricity grid; however this option is limited due to the complex terrain and dispersed nature of rural villages in some regions. Under these circumstances, stand-alone electrification systems that use renewable energy sources are a suitable alternative for providing electricity to isolated communities (Chaureya et al., 2004; El Bassam, 2001). The main advantages of these systems are that they are often much cheaper than grid extension and promote the long-term sustainability of the projects because of the use of local resources and avoidance of external dependences.

Photovoltaic (PV) systems have been widely used in the last decade to electrify rural communities. Usually individual PV home systems are used to cover basic household needs. Wind systems have recently been receiving increased attention (Lew, 2000; Nfah and Ngundam, 2008; Ekren and Ekren, 2009; Ferrer-Martí et al., 2011a). In particular, wind power has recently been used in several electrification projects in Cajamarca, a region in the northern highlands of Peru. El Alumbre was the first wind electrification project in Peru (Ferrer-Martí et al., 2010), to use individual wind turbines for each household. In windy areas, the ratio investment/produced energy can significantly diminish with wind systems compared to PV systems. Thus, the cost can be reduced, especially when demand increases and more powerful wind turbines are used (for instance, when supplying to groups of households with microgrids). Moreover using microgrids has social advantages (Kirubi et al., 2009; Alzola et al., 2009). However, in scattered communities combinations of small microgrids and individual systems for isolated households may be the best option (Lambert and Hittle, 2000; Camino-Villacorta et al., 2010; Ferrer-Martí et al., 2011b).

Hybrid systems that use PV and wind generators are increasingly being used (Deshmukh and Deshmukh, 2008) and the analysis and comparison of the most appropriate energy sources has gained the attention of researchers during the last decades (Khalil, 1981; Georgopoulou et al., 1997; Cherni et al., 2007). In particular, at present, research on stand-alone electrification systems for rural villages is mainly focused on the study of the best combination of energy sources, depending on the availability of resources. Different methods have been used, and the depth of analysis of various aspects depends on the particular study (Bernal-Agustín and Dufo-Lopez, 2009). Some studies consider in detail the cost of the components and equipment throughout the lifespan of the project (Ashok, 2007; Bala and Siddique, 2009). Huang et al. (2008) presented a mixed integer linear programming (MILP) model to optimize a community system, and took into account the perspectives of the different actors and institutions involved in the project.

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Ekren and Ekren (2009) developed a simulation model for a wind–PV system already designed. HOMER by NREL is the most widely used decision support tool, which simulates and compares costs throughout the lifespan of a project for various electrification options (Akella et al., 2007).

The design of electric distribution grids has been widely studied under different considerations (Ponce de Leao and Matos, 1999; Avella et al., 2005), but has been less so within the context of developing countries. As White et al. (2011, p. 6) state: "In face of worldwide difficulties of meeting energy needs and rising usage in developing countries, a greater emphasis on optimal use of energy resources might be expected in OR literature". In particular, few studies have considered the design of the possible microgrids with energy storage in batteries for rural isolated communities. Ter-Gazarian and Kagan (1992) presented a MILP model that did not allow individual generation points, but took into account the energy and the power demand simultaneously, selected the wire to use in the grids depending on the voltage props and solved the location of the batteries. Keller et al. (2007) presented a simulation tool to design an approximated solution of a microgrid with storage elements, and included different quantitative criteria (for instance, cost of the energy or percentage of the energy demand that was not met) and qualitative criteria (for instance, inhabitants agreement or visual impact). VIPOR by NREL is a decision support tool and considers both individual generation points and a microgrid with limited possible generation points. VIPOR uses a simulated annealing heuristic approach to solve the problem and minimize costs throughout the lifespan of the project (Lambert and Hittle, 2000). A similar algorithm to Lambert and Hittle's has been used at regional scale to define which villages to connect to a national electric grid and which villages to electrify through isolated systems, depending on the population density (Parshall et al., 2009).

To our knowledge, the only exact model for designing isolated electrification projects using both individual systems and microgrids has recently been presented in Ferrer-Martí et al. (2011b) which developed a MILP model based on binary variables but only considered wind generation. Doubts may arise when considering if solving mathematical programming models may be efficient enough for real size problems. However, as Atamtürk and Savelsbergh (2005, p. 69) stated: "integer programming is rapidly gaining acceptance as a powerful computational tool that can provide optimal or near-optimal solutions to real-life strategic and operational planning problems". Thus, the development of new MILP models to approach new problems now has significant attention in scientific literature (for instance, Corominas et al., 2011). Recently, Fomin and Kratsch (2010) confirm that the development of the area of exact algorithms to solve problems changed drastically in the last decade and is now a dynamic research area that has already significantly improved the running time of classical problems. In particular, the model of Ferrer-Martí et al. (2011b) has effectively been used to design the electrification project in Alto Peru (Cajamarca, Peru) where only wind energy was considered.

This paper presents a MILP model and exact solving procedure to optimize the design of not only wind but hybrid wind–PV rural electrification systems using individual generators and one or more microgrids. The model takes into account the power and energy demand at the consumption points and the energy resource maps. Then, the model solves the location and size of the equipment and the design of the microgrids and selects the best generation option or combination for each point in the community. The objective is to minimize cost, as it is usually done in literature (Akella et al., 2007; Ashok, 2007), because the budget is generally a tight constraint in rural electrification projects. As including a new generation option complicates the model and may make the solving process more difficult, we decided to develop and test different formulations to identify and choose the more efficient one. We developed two alternative mathematical formulations for the model: one uses integer variables to define the location and size of the equipment whereas the other uses binary variables to be able to analyze their efficiency and select the best one. First, a computation experiment is carried out to compare the efficiency of both formulations, to select the best one and to analyze the efficiency of the hybrid model in solving real size problems. Following this, the results of applying the hybrid model to the real case studies of El Alumbre and Alto Peru (both in Peru) are presented and analyzed to quantify the cost reductions obtained with the hybrid model compared to the real projects. Although including an additional technology may complicate the modeling and the solution process, the results obtained show that the model presented in this paper can help institutions that promote electrification projects to easily and efficiently design hybrid wind–PV systems that combine the use of individual generators and microgrids. This is advantageous for rural electrification projects.

The rest of the paper is organized as follows. Section 2 presents the design considerations of the hybrid rural electrification system and the mathematical programming model. Section 3 presents a computational experiment to analyze the performance of the hybrid model and Section 4 describes the application of the model to case studies. Finally, Section 5 is devoted to conclusions and future works.

2. Mathematical programming model

In this section, we present the design considerations of the hybrid rural electrification system and the mathematical programming model that we developed for the design of the electrification system.

2.1. Design considerations

The scheme of the elements involved in a wind–PV electrification system is briefly presented in Fig. 1. Hybrid systems that combine these two energy technologies have the advantage of being able to use the best generation option at each point of the community or a combination of both technologies. In large and mountainous communities wind resource may significantly vary between different points, and the use of hybrid systems allows, for instance, the taking advantage of wind generation in windy areas whereas PV panels are used in points with low wind resource.

- *Consumption points:* Energy and power demands are defined specifically for each consumption point. At the consumption points that are fed by a microgrid, an electric meter is needed to limit the energy consumption of each user to fairly share the energy stored in the batteries between all the consumers.
- *Wind generation:* Wind turbines produce energy by transforming wind kinetic energy to mechanical energy and then to electrical energy. Different types of wind turbine generators generate different amounts of energy. Wind turbines generate alternating current (AC) and then a controller (which may, or may not, be included in the wind turbine) transforms it to direct current (DC) and controls the charge/discharge of the battery. The energy produced by a particular wind turbine type depends on its location and the wind resource there. The wind turbines can only be placed at the consumption points and there is a limit to the number of turbines that can be placed at the same point.
- *PV generation:* PV panels produce energy by transforming solar radiation to electrify by means photovoltaic cells. Different types of PV panels generate different amounts of energy. PV

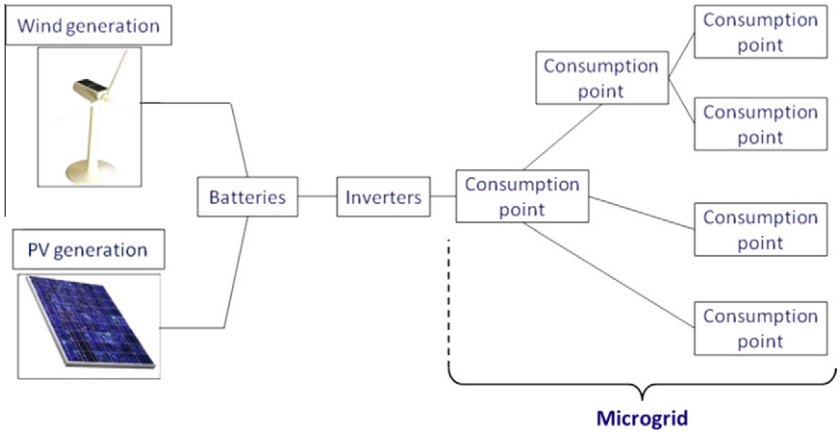


Fig. 1. Scheme of the elements involved in a hybrid wind–PV electrification system with microgrid distribution.

panels generate DC and then a controller ensures an appropriate charge/discharge of the battery. The energy produced by a particular PV panel does not depend on the location as the solar resource is considered uniform throughout the community. PV generators can only be placed at the consumption points and there is a limit to the number of PV generators that can be placed at the same point.

- *Energy storage:* The generated energy is first stored in a bank of batteries and then consumed from the batteries, which receive and supply electricity at DC. The capacity of the batteries is calculated according to the demand and the autonomy required. The bank of batteries can only be located at the generation points. In each bank of batteries there are inverters (DC/AC).
- *Microgrids*: Grid wires configure the microgrid that distributes the energy in a radial design. The total length of the microgrid is limited by the maximum voltage drop admitted; there is also a limit on the maximum admissible intensity in an electric wire. The power and energy losses in the grid wires are estimated according to the maximum voltage drop admitted.

2.2. Mathematical model

This paper presents a MILP model to optimize the design of hybrid wind–PV rural electrification systems using individual generators and one or more microgrids. The model takes into account the location and demand of each consumption point (power and energy demand) together with the energy resource maps (which defines the amount of daily energy that would be produced by each type of generator at each point of the map) to design the electrification system. Specifically, the model decides the location and size of the equipment and the design of the microgrids. The model solves the location and sizing of the equipment and the design of the microgrids to minimize the initial investment cost.

Therefore, there are two main differences between this model and the previous models found in literature (Ferrer-Martí et al., 2011b). First, this model considers not only wind generation but also PV generation and can design hybrid systems that select the best generation option or combination for each point in the community. The inclusion of a second generation technology makes the model more realistic and significantly increases the difficulty in solving it. In order to ensure the solving effectiveness of the problem, as the second main difference, we highlight that we developed two alternative mathematical formulations for the model to be able to analyze the performance of each one and select the best one. First, we present the integer model (IM) and then we summarize the changes needed to build the hybrid model using binary variables (BM). The main difference between both models is that the IM uses less variables but uses integer ones, which may get a range of numbers in the solution. The BM uses more variables but binary ones, which may only get two possible numbers in a solution. The effectiveness or superiority of these two possible formulations is difficult to know in advance, so we decided to test both models to finally choose the best one. The performance of both models is compared in Section 3.

2.2.1. Data

Consumption points

P	Number of consumption points (which may be, for instance, households, schools, health centers or community centers). These are the points where the generators can be placed
L_{pd}	Distance [meter] between two points p and d ($p = 1,\ldots,P$; $d = 1,\ldots,P$)
L_{max}	Maximum length of segment of a wire of the microgrid
Q_p	Set of points d to which a point p could be directly joined with a wire segment ($p = 1,\ldots,P$; $d = 1,\ldots,P$: $p \neq d$, $L_{pd} \leqslant L_{max}$)
ED_p	Electric energy demand [Watt hour/day] at p ($p = 1,\ldots,P$)
PD_p	Power demand [Watt] at p ($p = 1,\ldots,P$)
CM	Cost [US dollar] of an electric meter

Wind generation

A, NA	Types of wind turbines ($a = 1,\ldots,A$) and maximum number of wind turbines that can be placed at a point, respectively
EA_{pa}	Energy generated [Watt hour/day] by a wind turbine placed at point p of the type a ($p = 1,\ldots,P$; $a = 1,\ldots,A$)
PA_a	Maximum power [Watt] of a wind turbine of type a ($a = 1,\ldots,A$)
CA_a	Cost [US dollar] of a wind turbine of type a ($a = 1,\ldots,A$)
R	Types of battery charge wind controllers ($r = 1,\ldots,R$)
PR_r	Maximum power [Watt] of a battery charge wind controller of type r ($r = 1,\ldots,R$)
CR_r	Cost [US dollar] of a battery charge wind controller of type r ($r = 1,\ldots,R$)

(*continued on next page*)

PV generation

S, NS	Types of PV panels ($s = 1,\ldots,S$) and maximum number of PV panels that can be placed at a point, respectively
ES_s,	Energy generated [Watt hour/day] by a PV panel of the type s ($s = 1,\ldots,S$)
PS_s	Maximum power [Watt] of a PV panel of type s ($s = 1,\ldots,S$)
CS_s	Cost [US dollar] of a PV panel of type s ($s = 1,\ldots,S$)
Z	Types of PV battery charge controllers ($z = 1,\ldots,Z$)
PZ_z	Maximum power [Watt] of a battery charge PV controller of type z ($z = 1,\ldots,Z$)
CZ_z	Cost [US dollar] of a battery charge PV controller of type z ($z = 1,\ldots,Z$)

Energy storage

B	Types of batteries ($b = 1,\ldots,B$)
ηb	Efficiency of the batteries [fraction of unity]
DB	Maximum proportion of discharge admitted in the batteries
VB	Required autonomy of the batteries [days]
EB_b	Capacity [Watt hour] of a battery of type b ($b = 1,\ldots,B$)
CB_b	Cost [US dollar] of a battery of type b ($b = 1,\ldots,B$)
I	Types of inverters ($i = 1,\ldots,I$)
ηi	Efficiency of the inverters [fraction of unity]
PI_i	Maximum power [Watt] of an inverter of type I($i = 1,\ldots,I$)
CI_i	Cost [US dollar] of an inverter of type I($i = 1,\ldots,I$)

Microgrid

C	Types of microgrid wires
RC_c	Electric resistance (feed and return) [Ohm/meter] of a wire of type c ($c = 1,\ldots,C$)
IC_c	Maximum intensity [Ampere] of a wire of type c ($c = 1,\ldots,C$)
CC_c	Cost [US dollar/meter] of a wire of type c (feed and return), including the cost of infrastructure ($c = 1,\ldots,C$)
V_N, V_{min}, V_{max}	Nominal, Minimum and Maximum voltage [volt], respectively
ηc	Efficiency of the microgrid [fraction of unity]

2.2.2. Integer model (IM)

2.2.2.1. Variables. The IM uses the following types of variables:

- Integer non-negative variables to define the location and sizing of equipment
  - xa_{pa} Number of wind turbines of type a is placed at point p ($p = 1,\ldots,P$; $a = 1,\ldots,A$).
  - xs_{ps} Number of PV panels of type s is placed at point p ($p = 1,\ldots,P$; $s = 1,\ldots,S$).
  - xb_{pb} Number of batteries of the type b is placed at point p($p = 1,\ldots,P$; $b = 1,\ldots,B$).
  - xr_{pr} Number of battery charge wind controllers of type r is placed at point p($p = 1,\ldots,P$; $r = 1,\ldots,R$).
  - xi_{pi} Number of inverters of the type i is placed at point p($p = 1,\ldots,P$; $i = 1,\ldots,I$).
  - xz_{pz} Number of battery charge PV controllers of type z is placed at point p($p = 1,\ldots,P$; $z = 1,\ldots,Z$).
- Float non-negative variables to define energy and power flows and voltage
  - fe_{pd} Flow of energy [Watt hour/day] between the points p and d ($p = 1,\ldots,P$; $d \in Q_p$).
  - fp_{pd} Flow of power [Watt] between the points p and d ($p = 1,\ldots,P$; $d \in Q_p$).
  - v_p Voltage [volt] at the point p ($v_p = V_{min},\ldots,V_{max}$; $p = 1,\ldots,P$).
- Binary variables to define the generation points, the microgrid and the meters
  - $xg_p \in \{0;1\}$ 1, if some wind turbine and/or PV panel is placed at point p($p = 1,\ldots,P$).
  - $xc_{pdc} \in \{0,1\}$ 1, if there is a wire of type c between the points p and d ($p = 1,\ldots,P$; $d \in Q_p$; $c = 1,\ldots,C$).
  - $xm_p \in \{0,1\}$ 1, if an electric meter is placed at point p ($p = 1,\ldots,P$).

2.2.2.2. Objective function. The objective function (1) minimizes the initial investment costs, considering the generation, storage and distribution equipment.

$$[MIN]Z = \sum_{p=1}^{P}\sum_{a=1}^{A} CA_a \cdot xa_{pa} + \sum_{p=1}^{P}\sum_{s=1}^{S} CS_s \cdot xs_{ps} + \sum_{p=1}^{P}\sum_{b=1}^{B} CB_b \cdot xb_{pb} + \sum_{p=1}^{P}\sum_{d\in Q_p}\sum_{c=1}^{C} L_{pd} \cdot CC_c \cdot xc_{pdc} + \sum_{p=1}^{P}\sum_{i=1}^{I} CI_i \cdot xi_{pi} + \sum_{p=1}^{P}\sum_{r=1}^{R} CR_r \cdot xr_{pr} + \sum_{p=1}^{P}\sum_{z=1}^{Z} CZ_z \cdot xz_{pz} + \sum_{p=1}^{P} CM \cdot xm_p \quad (1)$$

2.2.2.3. Constraints. Constraint (2) defines the points at which wind turbines are placed and limits the maximum number of generators at the same point; in an analogous way (3) incorporates PV panels. Constraint (4) forces xg_p to be equal to 0 if neither a wind turbine nor a PV panel is placed at point p. Energy and power balances and conservation are described in (5) and (6), respectively. Constraint (7) establishes the capacity of the batteries. Constraints (8) and (9) relate the energy and power flows respectively, to the existence of a wire between two points. The radial distribution of the microgrid is established in (10), constraint (11) limits the voltage drops and (12) the maximum intensity. The power of battery charge wind controllers is defined in (13). In a similar way, the power of battery charge solar controllers is defined depending on the power of the corresponding PV panel (14). Inverters can only be placed at points where wind–PV generators are placed (15). Constraints (16) and (17) force the installation of electric meters at the consumption points fed by a microgrid.

$$\sum_{a=1}^{A} xa_{pa} \leqslant NA \cdot xg_p \quad p = 1,\ldots,P \quad (2)$$

$$\sum_{s=1}^{S} xs_{ps} \leqslant NS \cdot xg_p \quad p = 1,\ldots,P \quad (3)$$

$$\sum_{a=1}^{A} xa_{pa} + \sum_{s=1}^{S} xs_{ps} \geqslant xg_p \quad p = 1,\ldots,P \quad (4)$$

$$\sum_{q=1|p\in Q_q}^{P} fe_{qp} + \sum_{a=1}^{A} EA_{pa} \cdot xa_{pa} + \sum_{s=1}^{S} ES_s \cdot xs_{ps} \geqslant \frac{ED_p}{\eta b \cdot \eta i}\left(\frac{1}{\eta c} + \left(1 - \frac{1}{\eta c}\right)xg_p\right) + \sum_{d\in Q_p} fe_{pd} \quad p = 1,\ldots,P \quad (5)$$

$$\sum_{q=1|p\in Q_q}^{P} fp_{qp} + \sum_{i=1}^{I} PI_i \cdot xi_{pi} \geqslant PD_p\left(\frac{1}{\eta c} + \left(1 - \frac{1}{\eta c}\right)xg_p\right) + \sum_{d\in Q_p} fp_{pd} \quad p = 1,\ldots,P \quad (6)$$

$$\sum_{b=1}^{B} EB_b \cdot xb_{pb} + \left(\frac{VB}{DB}\sum_{j=1}^{P}\frac{ED_j}{\eta b \cdot \eta i \cdot \eta c}\right)(1 - xg_p) \geqslant \frac{VB}{DB}\left(\sum_{d\in Q_p} fe_{pd} + ED_p\right) \quad p = 1,\ldots,P \tag{7}$$

$$fe_{pd} \leqslant \left(\sum_{j=1}^{P}\frac{ED_j}{\eta b \cdot \eta i \cdot \eta c}\right)\sum_{c=1}^{C} xc_{pdc} \quad p = 1,\ldots,P; \quad d \in Q_p \tag{8}$$

$$fp_{pd} \leqslant \left(\sum_{j=1}^{P}\frac{PD_j}{\eta c}\right)\sum_{c=1}^{C} xc_{pdc} \quad p = 1,\ldots,P; \quad d \in Q_p \tag{9}$$

$$\sum_{q=1|p\in Q_q}^{P}\sum_{c=1}^{C} xc_{qpc} + xg_p \leqslant 1 \quad p = 1,\ldots,P \tag{10}$$

$$v_p - v_d \geqslant \frac{L_{pd} \cdot RC_c \cdot fp_{pd}}{V_n} - (V_{max} - V_{min})(1 - xc_{pdc}) \quad p = 1,\ldots,P; \quad d \in Q_p; \quad c = 1,\ldots,C \tag{11}$$

$$\frac{fp_{pd}}{V_n} - \left(\sum_{j=1}^{P}\frac{PD_j}{V_{min} \cdot \eta c}\right)(1 - xc_{pdc}) \leqslant IC_c \quad p = 1,\ldots,P; \quad d \in Q_p; \quad c = 1,\ldots,C \tag{12}$$

$$\sum_{r=1}^{R} PR_r \cdot xr_{pr} \geqslant \sum_{a=1}^{A} PA_a \cdot xa_{pa} \quad p = 1,\ldots,P \tag{13}$$

$$\sum_{z=1}^{Z} PZ_z \cdot xz_{pz} \geqslant \sum_{s=1}^{S} PS_s \cdot xs_{ps} \quad p = 1,\ldots,P \tag{14}$$

$$xi_{pi} \leqslant NI \cdot xg_p \quad p = 1,\ldots,P; \quad i = 1,\ldots,I \tag{15}$$

$$\sum_{d\in Q_p}\sum_{c=1}^{C} xc_{pdc} \leqslant (P-1)xm_p \quad p = 1,\ldots,P \tag{16}$$

$$\sum_{q=1|p\in Q_q}^{P}\sum_{c=1}^{C} xc_{qpc} \leqslant xm_p \quad p = 1,\ldots,P \tag{17}$$

2.2.3. Binary model (BM)

The main difference between the IM model and the BM model is the use of binary instead of integer variables to define the location and sizing of wind turbines, PV panels, batteries, controllers and inverters. Changes between both models only refer to the definition of variables: in the IM there is one variable to define the number of units of one type of equipment placed at one point whereas the BM uses a set of binary variables to define the number of identical units. Therefore, constraints of the BM are the same as the IM but the integer variables in the integer model must be replaced by the sum of the corresponding binary variables.

xa_{pan} 1, if the wind turbine number n of type a is placed at point p ($p = 1,\ldots,P$; $a = 1,\ldots,A$; $n = 1,\ldots,NA$)
xs_{psn} 1, if the PV panel number n of type s is placed at point p ($p = 1,\ldots,P$; $s = 1,\ldots,S$; $n = 1,\ldots,NS$)
xb_{pbn} 1, if the battery number n of the type b is placed at point p ($p = 1,\ldots,P$; $b = 1,\ldots,B$; $n = 1,\ldots,NB$)
xr_{prn} 1, if the battery charge controller number n of the type r is placed at point p ($p = 1,\ldots,P$; $r = 1,\ldots,R$; $n = 1,\ldots,NR$)
xz_{pzn} 1, if the battery charge controller number n of the type z is placed at point p ($p = 1,\ldots,P$; $i = 1,\ldots,I$; $n = 1,\ldots,NZ$)
xi_{pin} 1, if the inverter number n of the type i is placed at point p ($p = 1,\ldots,P$; $i = 1,\ldots,I$; $n = 1,\ldots,NI$)

The formulation of the BM model can be straightforwardly obtained from the IM model by replacing each integer variable (for example, xa_{pa}) in the objective function and the constraints by the sum of the corresponding binary variables (for example, $\sum_{n=1}^{NA} xa_{pan}$). Thus, for example, the equivalent constraint (2) of the BM model would be: $\sum_{a=1}^{A}\sum_{n=1}^{NA} xa_{pan} \leqslant NA \cdot xg_p, p = 1,\ldots,P$. Moreover, in the BM additional constraints are included to avoid symmetries, as has been done in a previous related work (Ferrer-Martí et al., 2011b).

3. Computational experiment

In the previous section two alternative mathematical models have been presented to the design of hybrid wind–PV rural electrification systems using individual generators and one or more microgrids. The main difference between both models is that they use integer or binary variables to define the location and size of equipment. In this section, we analyze the effectiveness of these models as a decision making tool for institutions (NGOs, enterprises, local governments) that promote electrification projects. The models can help them to easily design hybrid wind–PV systems with microgrids.

To analyze the performance of the models, we carried out a computational experiment in which the instances were randomly generated based on the characteristics of the real projects in terms of number of consumption points, resource maps, energy and power demand and components. The real projects considered were El Alumbre and Alto Peru, which are described in more detail in Section 4. The available computing time was defined according to potential future project designers for which a model like the one presented in this paper may aid the design process.

The main characteristics of these instances are:

- *Communities*: Community (1) the area (3.5 kilometer × 3.5 kilometer) and the wind map are taken from El Alumbre (Ferrer-Martí et al., 2011b). Community (2) the area (1.3 kilometer × 3.4 kilometer) and the wind map are taken from Alto Peru. The wind resource maps are calculated using WAsP (Mortensen and Peterson, 1997). Wind resource in both communities is variable, there are parts of the community where the wind resource is high whereas in other parts it is low-to moderate. Solar resource is constant and equal for both communities; Peak Solar Hour is 4.3.
- *Number households:* 5,10,15,…,50.
- *Concentration*: (1) households are randomly dispersed, (2) 25% of the households are concentrated in 30% of the area, and (3) 50% of the households are concentrated in 30% of the area.
- *Demand (energy/power) per household: (1) 280 Watt hour/day, 200 Watt and (2) 420 Watt hour/day, 300 Watt.*
- *Equipment*: Technical data and cost of generators (wind turbines, PV panels) and also of the other equipment (wind and PV controllers, inverters, wires, meters) including the types that were used in the real projects and more powerful devices. Wind turbines (four types): nominal power: 100 Watt, 500 Watt, 1000 Watt and 2000 Watt; maximum power: 300–3500 Watt; cost: US dollar 974–US dollar 5132. Wind controllers (four types): maximum power: 420–3600 Watt; cost: US dollar 165 to US dollar 513. PV panels (four types) 210 Watt hour/day to

629 Watt hour/day, cost from US dollar 451 to US dollar 1000. PV controller (four types), maximum power from 50 Watt to 200 Watt, cost from US dollar 67 to 1 US dollar 25. Batteries (four types): capacity: 1500–3000 Watt hour; cost: US dollar 225–US dollar 325; efficiency 85%. Inverters (four types): maximum power: 300–3000 Watt; cost: US dollar 377–US dollar 2700; efficiency 85%. Grid wires (two types): costs US dollar 4.9/meter–US dollar 5.1/meter; resistance: 2.71–2.1 Ohm/kilometer.

For each combination of the first parameters (communities, *COM*; number of households, *HOU*; concentration, *CON*; demand, *DEM*), five instances were generated, giving a total of 600 instances (so, 60 for each value of *HOU*, 200 for each value of *CON*, 300 for each value of *DEM*). The maximum calculation time was set to 18,000 seconds. Mathematical programs were solved using the IBM ILOG CPLEX 12.2 Optimizer, on a PC 3.16 GHz Intel Core 2 Duo E8500 with 3.46 GB of RAM. Table 1 summarises the results of the computational experiment for the integer and the binary models, depending on the characteristics of the instances, showing: the number of instances with a proved optimal solution (*Opt*), the average time ($\overline{Time}$) in seconds to obtain these optimal solutions and, for the rest of the instances, the GAP (%) that indicates the related difference between the values of the solution (Z) and the lower bound (LB), calculated as $\frac{100 \cdot (Z-LB)}{Z}$.

The results show the superiority of the integer model compared to the binary model; the IM obtains 9.3% more optimal solutions than the BM (469 vs. 429). Moreover, the average computing time to obtain the optimal solution is 43.4% smaller for the IM than for the BM (433.64 seconds vs. 766.33 seconds). The integer model is on average superior to the binary model in terms of number of optimal solutions and average computing time for all communities (*COM*), concentrations (*CON*), demands (*DEM*) and number of households (*HOU*), except for 40 households.

In addition, we have analyzed and compared the performances of the integer and the binary models when both models guarantee the optimal solution or when none of the models guarantees the optimality of the solution. When both models guarantee the optimal solution (428 instances), in almost all the instances the IM is quicker than the binary one (425 vs. 3 instances). When none of the models obtained a proved optimal solution (130 instances), solutions obtained are very similar (BM obtains solutions up to 0.3% greater than IM, depending on the characteristics). However, the IM finds a better solution in 74.6% of the instances (97) and the same solution in 4.6% of the instances (6); BM only finds a better solution in 20.8% of the instances (27). This superiority of the integer model is again shown for all levels of number of households, communities, concentrations and demand.

From now on we will focus on the integer model, which has been revealed to be the best one. The results of the IM model are very satisfactory: IM always finds a proved optimal solution in instances of up to 25 households. Moreover the optimal solutions are found in a very short time: even for instances of 25 households only an average of 174.81 seconds are needed. These instances are representative of the average size of the electrification projects (for example, the project in Alto Peru included 26 households). For greater instances up to 40, the optimal solution is found in most of the instances with less than 1600 seconds computing time, and for the rest of the instances GAPs are very low, only up to 3.8% in average. Even in instances of up to 50 households, GAPs are very low, up to 6.5% in average. In general, results are especially satisfactory from the point of view of real projects and future users: optimal solutions are guaranteed in a very short time for average sized usual projects and very close to optimal solutions are expected for the largest projects.

Table 1
Results of the computational experiment.

Total		IM			BM		
		Opt	$\overline{Time}$	GAP	*Opt*	$\overline{Time}$	GAP
		469	433.64	4.7	429	766.33	5.9
HOU	5	60	0.02	0	60	0.39	0
	10	60	0.23	0	60	7.80	0
	15	60	1.28	0	60	37.44	0
	20	60	32.84	0	60	234.10	0
	25	60	174.81	0	55	1154.36	0
	30	52	581.72	2.4	44	1410.93	3.8
	35	45	861.12	3.4	38	1818.00	5.2
	40	33	1589.46	3.8	24	1508.05	4.9
	45	23	1228.33	5.2	18	1414.75	6.9
	50	16	2570.06	6.5	10	5565.87	8.3
COM	1	273	346.65	3.8	259	678.86	3.9
	2	196	554.80	4.9	170	899.59	6.6
CON	1	168	344.93	3.6	151	621.94	4.8
	2	165	414.37	4.3	154	957.50	6.1
	3	136	566.59	5.5	124	704.74	6.5
DEM	1	251	420.06	4.1	239	736.57	4.7
	2	218	390.20	5.1	190	638.98	6.6

4. Case studies

In this section, the proposed hybrid wind–PV model is applied to the real cases of El Alumbre and Alto Peru (both in Peru) that were promoted by the NGOs Practical Action from Peru, Engineering Without Borders from Catalonia in Spain and Green Empowerment from the USA. The projects were financed through funds obtained from the three NGOs from different private and public institutions. Both projects use wind systems but in El Alumbre the electrification is based in individual systems installed at each consumption point whereas in Alto Peru the electrification system uses microgrids. El Alumbre was the first community wind electrification project in Peru; in 2008, 35 individual wind turbines were installed to electrify 33 households, the school and the health center. The next wind project was Alto Peru; the project was designed using the wind model of Ferrer-Martí et al. (2011b). The application of the designed hybrid model in these projects will evaluate the contribution of using this hybrid model with real projects that used only wind systems, either individual (El Alumbre) or with microgrids (Alto Peru).

We consider the real location of the households in both communities: in El Alumbre 33 households and a school and a health center with high demand; in Alto Peru we consider the 26 households in the area defined in the previous section. Table 2 compares the solutions (Cost, Solution Scheme) obtained using the new hybrid model (Hybrid model) proposed in this paper with the real projects.

The hybrid model presented in this paper can design hybrid systems that select the best generation option or combination for each point of the community. Therefore, the solutions obtained may use only one technology at one point whereas in another may use hybrid generation systems (according to demand, resource maps and considering all the technical equipment and constraints). The solutions presented in Table 2 show these possibilities. The solution for El Alumbre is a combination of technologies (all the microgrids and individual systems use only technology, wind or PV systems). In contrast, the solution for Alto Peru uses one hybrid microgrid and then other microgrids and individual systems fed by only PV generators.

Table 2 shows that in both communities, the solutions obtained with the new model use hybrid systems and microgrids. The solu-

Table 2
Comparison of the solutions of hybrid model and the real projects.

Solution	Hybrid model	Real project
El Alumbre		
Cost (US dollar)	58,674	75,517
Solution	1 Wind microgrid (11 users)	35 Wind individual users
Scheme	22 Wind individual users 2 PV individual users	
Alto Peru		
Cost (US dollar)	35,843	51,691
Solution	1 Wind–PV microgrid (13 users)	6 Wind microgrids (8, 6, 3, 3, 3, and 2 users)
Scheme	2 PV microgrids (2 and 2 users) 9 PV individual users	1 Wind individual user

tions obtained with the hybrid model significantly reduce the cost of the real projects. In El Alumbre, the cost is reduced by 22.3% (US dollar 58,674 vs. US dollar 75,517) and the percentage cost reduction in Alto Peru is even greater, 30.7% (US dollar 35,843 vs. US dollar 51,691). These cost reductions are obtained because the model can explore and select the optimal combination of microgrids systems, individual systems and for each one decide the optimal energy technology: wind turbines, PV panels or an hybrid wind–PV system. Finally, it is worth pointing out the efficiency of the new hybrid model as a tool to design real projects.

5. Conclusions and future prospects

This paper presents a MILP model for optimizing the design of rural hybrid wind–PV electrification systems. The solutions define the location of the generation points in the village, the microgrid design and the location and size of the other equipment involved in the projects to minimize the initial investment cost.

We developed two alternative mathematical formulations that use either integer or binary variables to define the location and size of the equipment. A computational experiment is carried out to compare the performance of the models; results show the superiority of the integer model and the efficiency of the hybrid model as a tool to assist promoters in solving real size instances for future projects. The application of the hybrid model in two real case studies in Peru confirms that the solutions obtained with the new model allow very significant cost reductions, as it can explore and select the optimal combination of microgrids, individual systems and the energy source that fed each one.

The hybrid model presented in this paper can help institutions that promote electrification projects (governments, enterprises, NGOs) to design efficient projects faster. For future research we are working on including the microhydraulic energy in the hybrid model; this technology has a special feature that generates continuously in comparison with wind and PV systems. Moreover, it would be very interesting to complement the objective of cost minimization with other solving criteria, for instance, in ease of management and maintenance of the system or environmental impact.

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